



Behavioral and neural signatures of representational relevance for object concepts

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Introduction

- Humans encounter many objects in everyday vision, but what is most frequent in the world is not necessarily what is most central in our minds.
- For vision science, this raises the question of how to quantify which object concepts are most representative or behaviorally relevant in human representations.
- Existing variables (e.g., age of acquisition, word frequency, prevalence, lexical decision times, typicality, semantic richness) relate to accessibility, but none was designed to capture this directly.
- We propose **Representational Relevance (RR)** as a construct of how centrally an object concept is represented and how likely it is to come to mind across people.

Aim of this work: Operationalize and validate RR as a measure of object-concept centrality and test its links to neural responses to object images.

Methods: Behavioral

Word Ranking Task:

- Online drag-and-drop experiment (Fig 1).
- Stimuli: 200 unambiguous THINGS nouns
- Instruction: "Imagine 1,000 people named a real thing – rank words by how often they would be named."
- Response: Rank 1–8 (frequently→rarely)
- N = 6,000 trials (~240 rankings/word)
- Analysis: RR = mean word rank; split-half reliability and correlations with psycholinguistic norms.

Example first level scale of the task:

frequently named **bracelet** rarely named

pen hotdog cardboard zebra compost pennant seismograph

☐ I do not know this word

Next Trial

Figure 2. Example first level scale of the task

Example second level scale of the task:

frequently named **bracelet** rarely named

hairbrush cardboard magnet zebra tuxedo compost guardrail

Go back to Step 1 Next Trial

Figure 3. Example second level scale of the task

Representational Relevance Rating Task:

- Online two-step anchored rating task.
- Stimuli: 1,388 THINGS nouns (real-world objects); ~40 ratings/word.
- Instruction: "Imagine 1,000 people named a real thing – rate how often each word would be named."
- Response: First, coarse placement on a 7-anchor scale (Fig 2) → Next, zoomed-in refinement (Fig 3)
- Analysis: N = 55,520 trials, RR = mean global scale position per word; split-half reliability and correlations with psycholinguistic norms.

What the task looks like

- If you think a word was said more often (e.g., "sea lion"), place it higher in the list.
- If you think a word was said least (e.g., "humidifier"), place it near the bottom.
- Any word you don't know goes into Set aside (unknown) (e.g., "axolotl").

Available words

Rank the words

Frequently named

1 sea lion

2

3

4

5

6

7

8 humidifier

Rarely named

Set aside (unknown)

axolotl

Results: Step 1 - Word Ranking Task

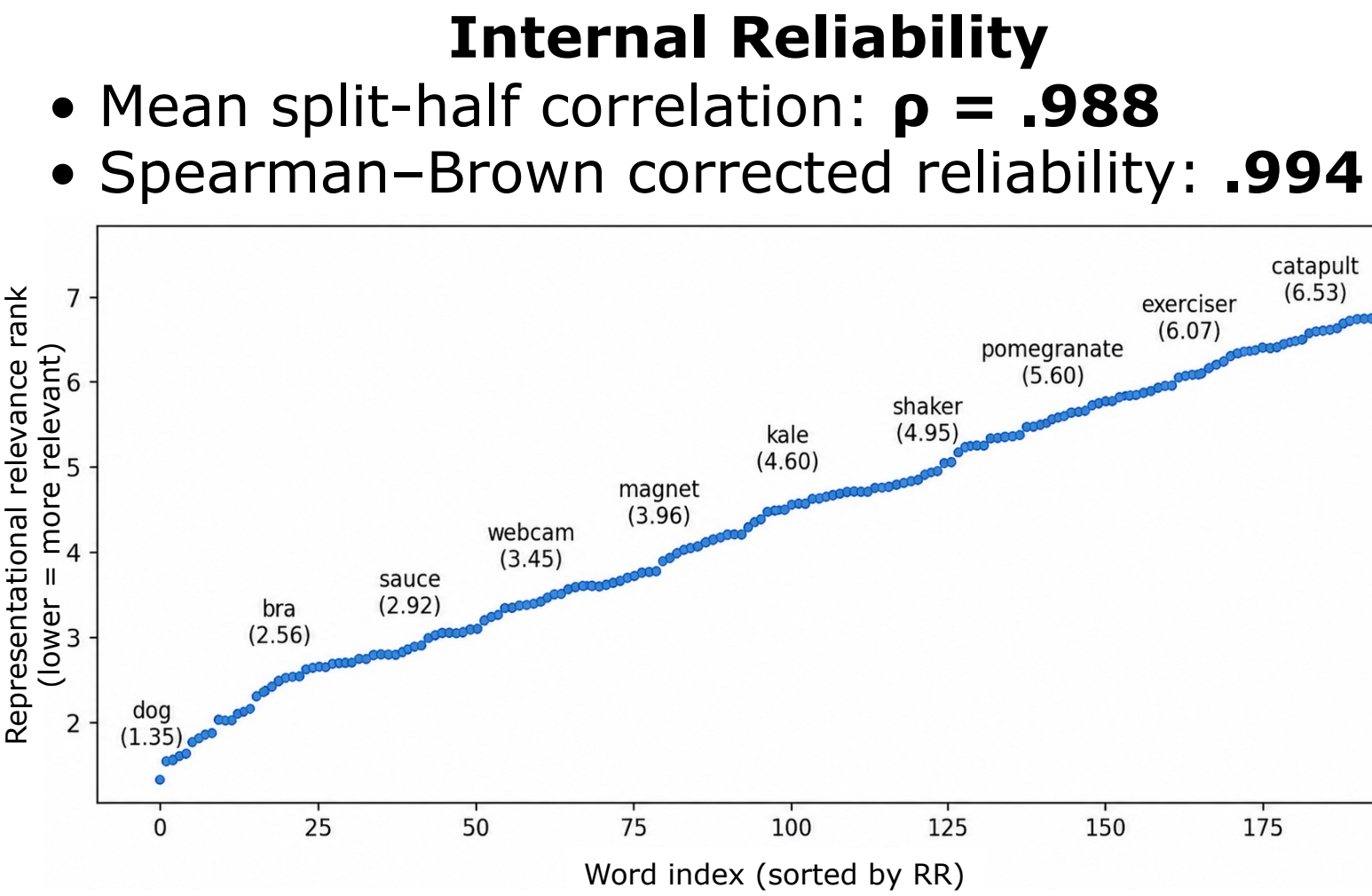


Figure 4. Sorted RR scores for the 200-word stimulus set, with every 20th item annotated. Lower values indicate higher representational relevance (more frequently named concepts).

External Validity

Table 1.

Correlations Between Word Ranking task RR Scores and Psycholinguistic Variables

Variable	Spearman ρ	Overlap (n)
AoA ^[1]	.753	194
LDT ^[2]	.734	200
Word Frequency ^[3]	-.660	200
Word Prevalence ^[4]	-.597	200
Semantic Richness ^[5]	-.282	200
Word Length	.323	200

Note. Word length refers to the number of letters. All correlations are significant at $p < .001$.

Results: Step 2 – Representational Relevance Rating

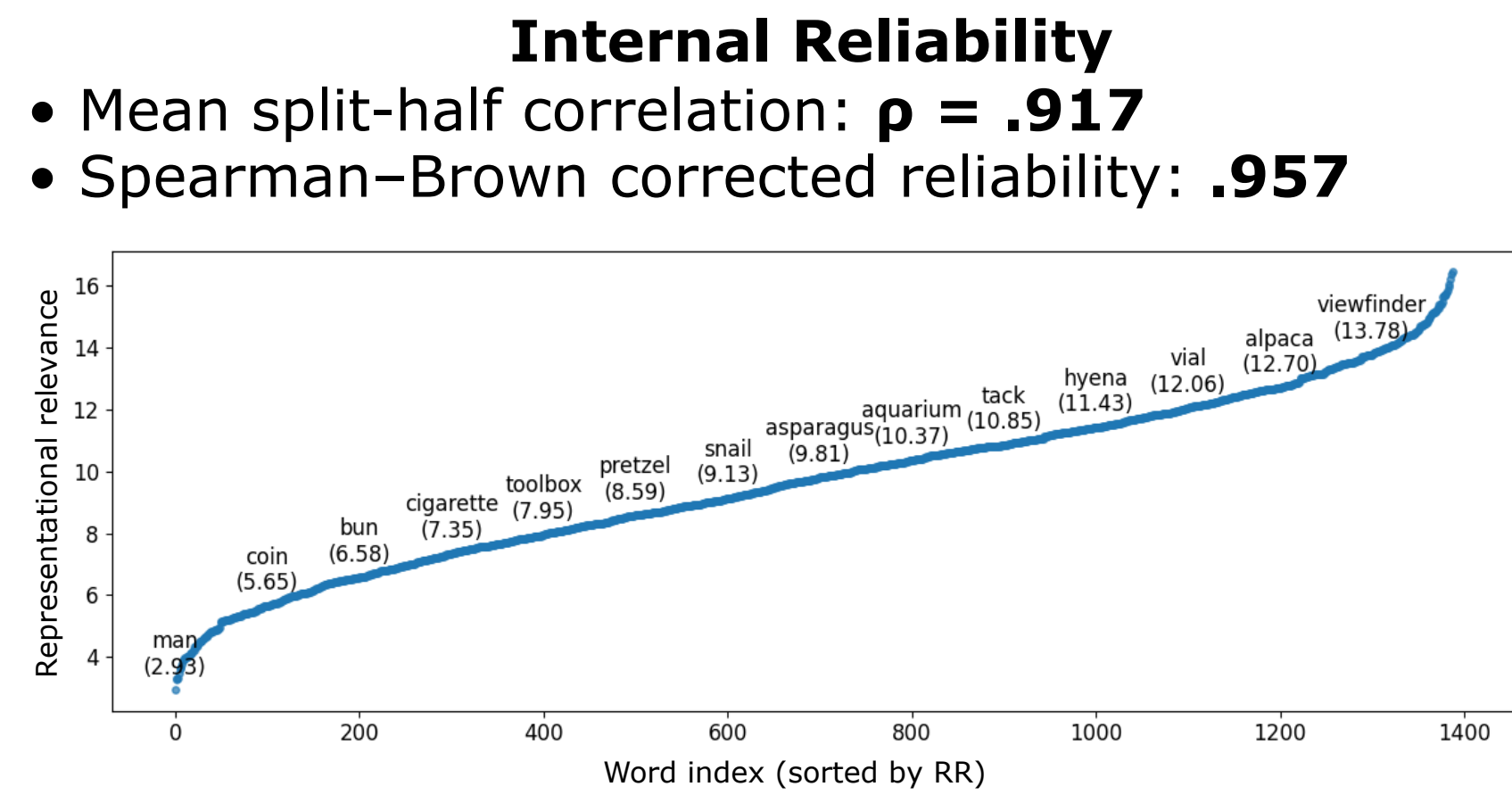


Figure 5. Sorted RR scores for the 1388-word stimulus set. Lower values indicate higher representational relevance (more frequently named concepts).

External Validity

Table 2.

Correlations Between Concept Rating task RR Scores and Psycholinguistic Variables

Variable	Spearman ρ	Overlap (n)
LDT	.706	1387
AoA	.732	1339
Word Frequency	-.685	1387
Word Prevalence	-.558	1387
Semantic Richness	-.203	1388
Word Length	.339	1388

Note. Word length refers to the number of letters. All correlations are significant at $p < .001$.

Prototypicality vs RR

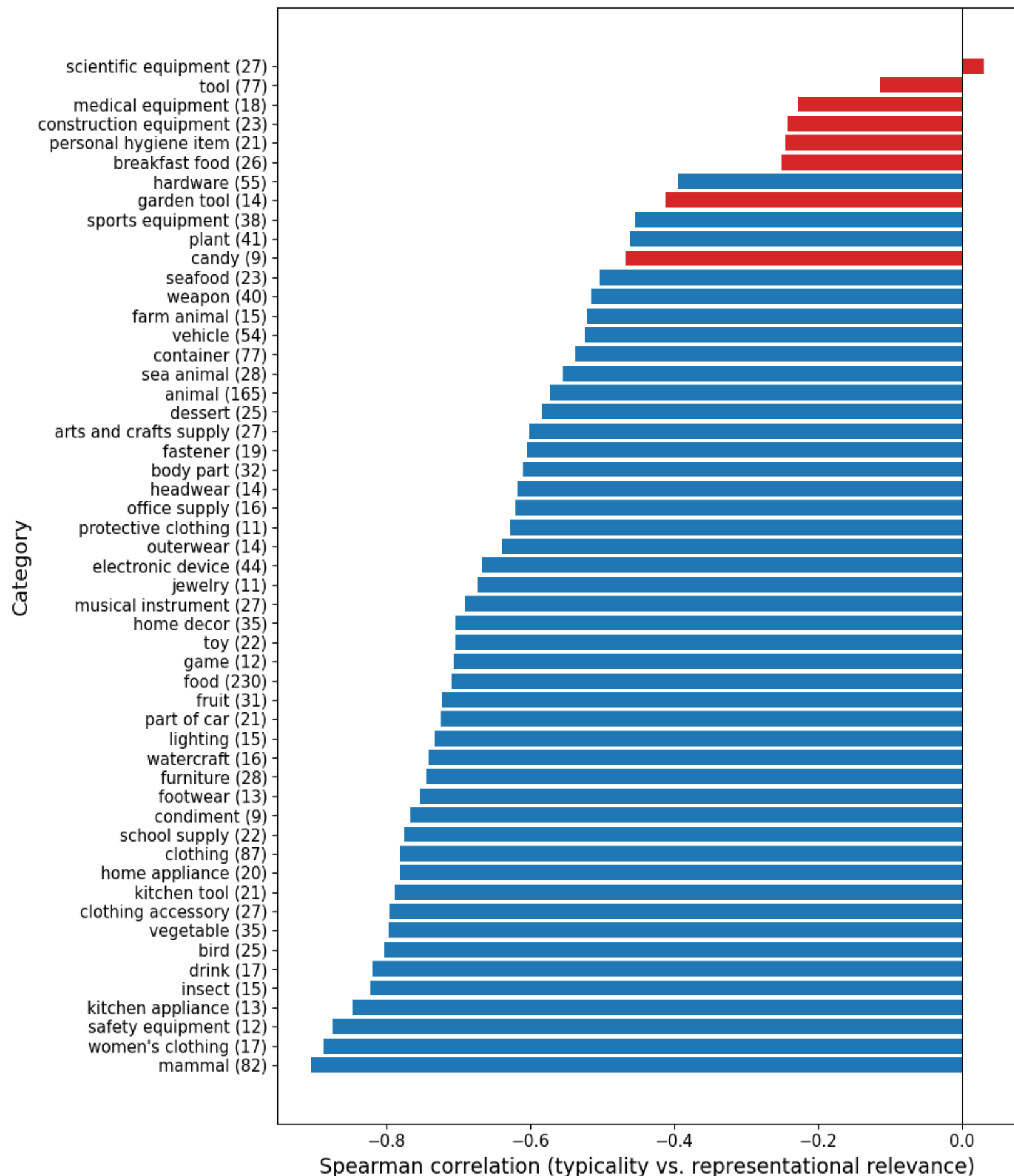


Figure 7. Category-Wise Correlations Between Prototypicality and RR. Significant correlations were displayed in blue bars ($p < .05$), and nonsignificant correlations were displayed in red bars. The number of items contributing to each estimate is shown in parentheses next to the category label.

Results: fMRI

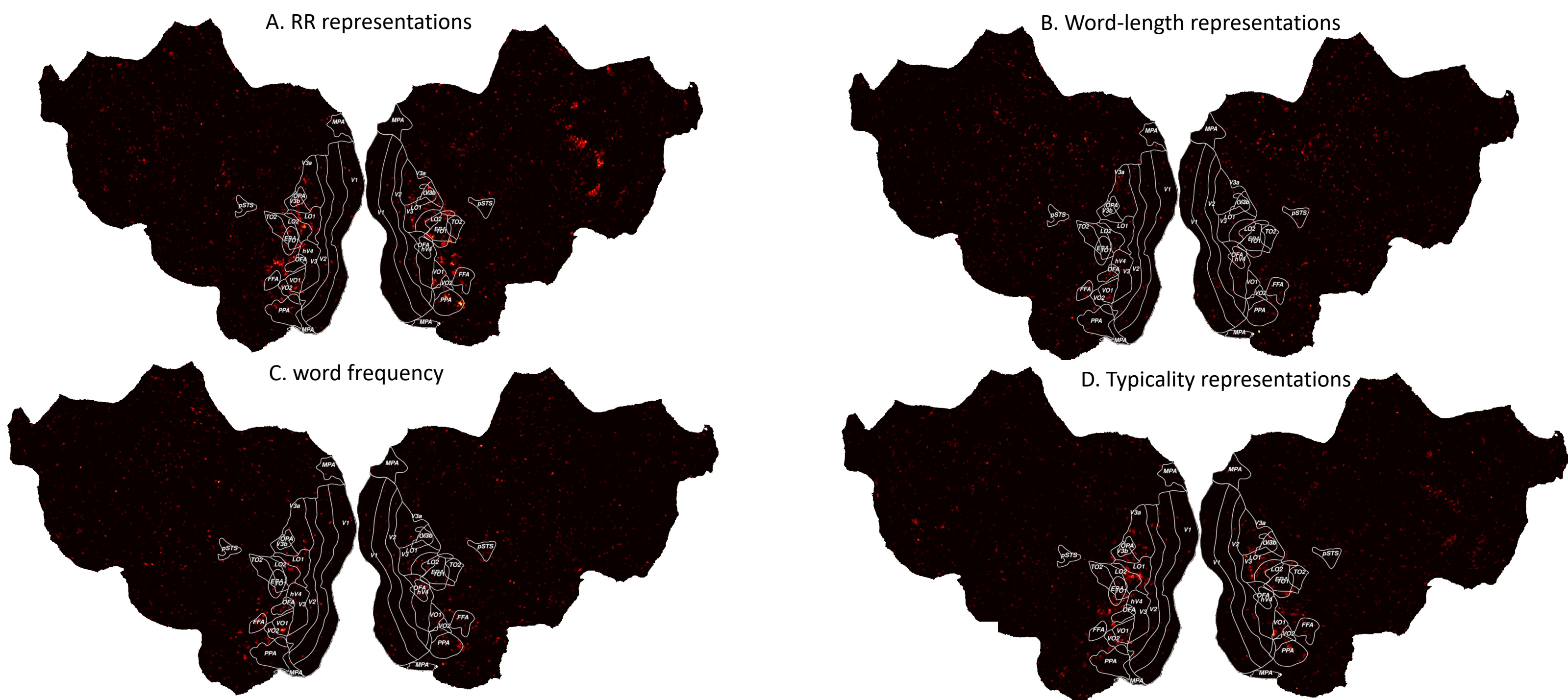


Figure 8. Cortical maps showing representations of RR, word length, word frequency, and typicality.

Discussion

- The main aim of this study was to operationalize representational relevance (RR) as a measure of object-concept centrality.
- The proposed tasks produced stable and interpretable estimates of RR.
- RR was systematically related to established psycholinguistic variables, while remaining distinct from them.
- Prototypicality was strongly associated with RR within many categories, with weaker relationships in some domains.
- fMRI encoding analyses showed that RR reliably predicted neural responses in high-level occipitotemporal cortex.

RR captures a graded and behaviorally meaningful dimension of object-concept centrality that is evident at behavioral and neural levels

Outlook: This provides a foundation for extending RR from object concepts to the representational relevance of images.

References

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